

UTHM Industrial Visit

NEWSLETTER



Summary

During the visit to UTHM, we are introduced with the strategic roadmap for ICT and Enterprise Architecture (EA) development. The speaker explained about how the current framework is anchored by the UTHM ICT Strategic Plan (2020–2024) and the EA implementation timeline is scheduled to conclude by June 2026. Ultimately, the initiative aims to refine the ICT workforce while creating a dynamic digital system that effectively supports the university's academic and administrative goals.

The talk is then followed by an exclusive visit to UTHM's newly built data center. We were introduced to key infrastructure features, including efficient thermal management using hot and cold aisle containment and how they maintain and manage the data center. The visit ended with interactive QNA sessions, which allows us to gain a deeper understanding of the university's broader digital landscape.

REFLECTION TAN YI YA A23CS0187

Attending the industry talk and site visit at UTHM really helped me connect what I've learned in theory to how things actually work on the ground. Being among the first to tour the new data center was a highlight, and seeing the hot and cold aisle cooling system in person made the technical concepts much clearer. The whole experience left me feeling curious, excited and more aware of how planning, governance, and engineering come together to run a digital campus. The visit taught me that digital transformation is a challenging process that requires careful planning.

TECHNOLOGIES AND CHALLENGES

At UTHM, the team has taken some steps to keep their digital systems safe and reliable. They use two firewalls (Europe & China) alongside a Web Application Firewall (WAF) to protect their websites and databases from online threats. Inside the data center, gas is used instead of water for fire suppression and the servers are arranged in hot-aisle and cold-aisle setups to control airflow and keep equipment running smoothly.

However, UTHM stills faces challenges with network setup, IPv6 and fitting servers into racks. They rely on local backups instead of cloud hosting and while the Learning Management System (LMS) is centralized, other systems remain separate. Migration to new web-based platforms has taken years and some processes still use paper forms.

REFLECTION TEH RU QIAN A23CS0191

From this visit, I learned that managing a data center is not only about having strong security tools but also about handling challenges everyday. UTHM's use of dual firewalls, WAF and cooling and fire suppression shows the importance of protecting system from both digital and physical risks. At the same time, I realized that issues like network setup, IPv6 configuration and fitting servers into racks can create obstacles. This taught me that building and maintaining digital infrastructure requires both technical expertise and problem-solving skills to balance security, efficiency and usability.